Street View Image Recognition System: Divide and Conquer

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Abstract

In this paper, we describe our approach to the Street-View Character Recognition competition [1]. Our initial experiments using a standalone multi-head EfficientNet-B0 yielded suboptimal performance when generalizing length-variable digit sequences. To overcome these limitations, we transitioned toward a decoupled 'divide and conquer' strategy that separates localization from recognition. Our final architecture utilizes a YOLOv8-based detection framework as a precise region proposal mechanism to extract individual digit bounding boxes, followed by a fined-tuned EfficientNet-B0 classifier that performs high-fidelity inference on normalized digit crops. A key contribution of this work is the introduction of a dual-stage training procedure—Semantic-to-Spatial Transfer Learning—where a multi-class detector is refined into a class-agnostic locator to maximize digit recall. Experimental results demonstrate that this hybrid pipeline significantly enhances robustness against urban noise and scale variations, achieving a 91.49% test accuracy on the competition. Additionally, we evaluate and compare our different solutions' performance on the contest task analyzing the critical trade-offs between classification accuracy and the total number of model parameters.

1. Introduction

Recognizing multi-digit sequences in natural scenes is challenging due to perspective distortions and complex backgrounds. We address the Street-View competition [1] using a **divide-and-conquer** strategy that decomposes the task into precise localization and digit classification. To maximize computational efficiency, we propose a **hybrid two-stage pipeline** featuring three key innovations. First, we couple a YOLOv8 localization model with a specialized EfficientNet-B0 classifier, allowing independent optimization of each stage. Second, we employ **cross-task progressive training**, where the detector is initially trained with classification heads to learn semantic features before being fine-tuned exclusively for class-agnostic localization. Finally, to support this decoupled architecture, we implement a **custom dataset synthesis** routine that extracts localized digit crops from the competition images, enabling targeted training of the second-stage classifier.

2. Previous work

Initially, our research focused on two baseline architectures. First, we implemented an **EfficientNet-B0 Multi-Head** classifier designed to predict digit sequences globally. This approach failed to achieve competitive results (less than 50% test accuracy), mainly due to two fundamental limitations:

- **Sequence Length Ambiguity:** The multi-head architecture requires a fixed number of output slots, which poses significant challenges when the exact number of digits in an image is unknown, often leading to "phantom" detections or missing characters.
- **Spatial-Ordered Correspondence:** There was a significant difficulty in mapping specific digits to their correct relative positions within the sequence. Without an explicit localization mechanism, the model struggled to maintain the structural order – especially when digits were unevenly spaced or subjected to perspective distortion.

Subsequently, we evaluated a standard **YOLOv8** object detector configured for 10-class recognition. This improved spatial awareness (solving the problem of **Spatial-Ordered Correspondence**) and resulted in an overall classification improvement, reaching 89.73% test accuracy. The performance of the model is summarized in the confusion matrix shown in Figure 1. Although the results are decent, we still observe significant "leakage" into the background class (false negatives) and false alarms(false positives) which result in missing characters and phantom detections, respectively. Therefore, we should improve both the recall, for solving the problem of missing characters, and precision, for solving the problem of phantom detections. Note that if any of the mentioned problems occur, they will ruin a whole digit-sequence detection, so both metrics are equally

important. Additionally, me might also need to decrease the number of miss-classifications.



Figure 1. Validation results: Confusion Matrix for the preliminary 10-class YOLOv8 model. The consider number of false negatives – many instances of digit 1 were overlooked– and false positives motivated a focused model for detection only.

3. Preliminaries, Discussion and Intuition

Our final model was developed to address the dual challenges of localization precision and recall. In this section, we discuss the intuition behind our most significant procedural innovations and the improvements obtained:

3.1. Transition to a Decoupled Model

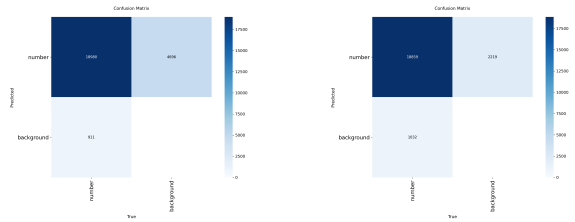
Our proposal centers on a modular network composed of two independent models with distinct objectives: a focused detector (**YOLOv8** with a single class head) and a specialized classifier (**EfficientNet-B0**). The core objective of this decoupling is to simplify the overall pipeline and to allow each architecture to specialize in the tasks they are most suitable for, thereby achieving better results, especially for detection. With this idea, although some validation precision was lost, we successfully improved the validation recall and final testing accuracy. We can see the results obtained in the confusion matrix shown in Figure 2a and in Table 1.

3.2. Semantic-to-Spatial Transfer Learning

The decision to first train a 10-head YOLOv8 model before transitioning to a single-head detector is grounded in the intuition that **semantic knowledge aids localization**.

By initially training the model to recognize 10 distinct digit classes, we force the backbone to learn rich, discriminative features (e.g., the specific curves of a '2' versus the verticality of a '1'). Once the model has internalized what constitutes a digit's identity, we perform a task-specific refinement: we collapse the output into a single-class head. This allows the network to re-purpose its semantic understanding to focus exclusively on detecting digits, essentially becoming an expert "digit locator".

Indeed, the empirical results, shown in Figure 2b and Table 1, proved the effectiveness of our idea in improving the overall result, achieving a balance between recall and precision.



(a) Normal (1-head) (b) Transfer (1-head)

Figure 2. Comparison of detection performance: "Normal" vs. "Transfer" training.

Table 1. Comparative performance analysis between the multi-class baseline and the proposed class-agnostic localization stage with the data of the confusion matrix(CM). "Tranfer" obtained an overall better performance.

Metric	**Baseline (10-Head)**	**Normal (1-Head)**	**Transfer (1-Head)**
True Positives (TP)	18,297	18,980	18,859
False Negatives (FN)	1,107	911	1,032
False Positives (FP)	2,300	4,696	2,219
Precision(CM)	88.83%	80.16%	**89.47%**
Recall(CM)	94.29%	**95.42%**	94.81%
Test Accuracy	89.73%	90.83%	**91.49%**

3.3. Data Amplification through Decoupling

The most significant advantage of our "Divide and Conquer" approach is the **multiplicative effect on the training set** for the classification stage.

In a standard holistic model, one input image corresponds to exactly one training sample (the full sequence). However, by decoupling the classification stage (EfficientNet-B0), we unlock a new level of data efficiency:

- **Sample Multiplication:** A single image containing, for example, a 5-digit sequence is transformed into five independent, high-quality training samples for the classifier (we passed from 30000 image samples to 65765 digit samples for training).
- **Input Normalization:** Cropping the bounding boxes allows the system to effectively remove noisy backgrounds and center relevant features.

This strategy provides the classifier with a more diverse and robust set of examples to learn from.

4. Model Architecture and Experiment

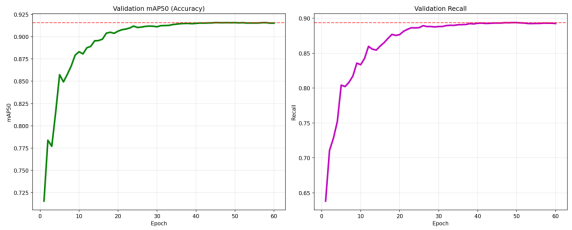
Now we are going to describe the architectural details of our network. It is formed by the following components:

4.1. Digit Detector (YOLOv8n)

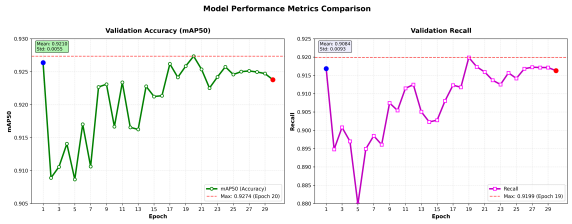
Although our framework is compatible with different networks, we selected the **YOLOv8 nano (YOLOv8n)** architecture for the detection stage due to its balance between performance and computational requirements. The detector underwent a two-phase training process:

1. **Semantic Pre-training:** Initially, we trained the network with a 10-class head, focused on learning discriminative features for each digit (0-9).
2. **Class-Agnostic Refinement:** Upon convergence, we replaced the 10-class head with a single-class ("digit") head and fine-tuned the model to focus exclusively on bounding box regression and objectness scores.

The training metrics obtained are shown in Figure 3.



(a) YOLO 10-head validation metrics. Best mAP50 (91.58%) and recall(89.38%)



(b) YOLO 1-head validation metrics (after transfer learning). Best mAP50 (92.74%) and recall (91.99%)

Figure 3. Validation metrics over training epochs. *Note that these values are averaged across multiple thresholds, explaining the variance from the fixed-threshold results in Table 1.

4.2. Data Preprocessing

Once the YOLOv8 detector identifies potential digits, the system extracts the regions of interest (RoIs) from the original image (crops of bounding boxes).

4.3. Specialized Classifier (EfficientNet-B0)

The second stage of our pipeline employs an **EfficientNet-B0** architecture, chosen for its efficiency and lightweight. The result model has a high-fidelity inference capacity, being able to distinguish between easily confused classes (e.g., '1' vs '7', or '5' vs '6'). As shown in Figure 4, the classifier

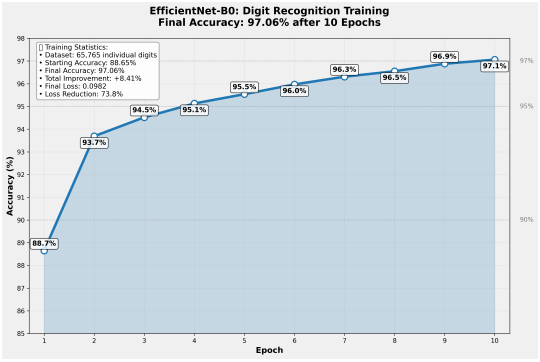


Figure 4. Efficient net training: The amount of data helped it to converge fast. Additionally, it reaches 97% of classification accuracy.

converges rapidly, reaching near-optimal accuracy within the first few epochs. This acceleration is attributed to the synthetic expansion of the dataset—training.

4.4. Configurations and details:

We present in Table 2 some hyper-parameters used for training. We note that this decoupling allowed a more stable and time-efficient training cycle, since, although the combined pipeline totals ∼7M parameters, we only had to train two *smaller* networks of ∼3.2M and ∼4M parameters.

Table 2. Specific training configurations for the YOLOv8 and EfficientNet-B0 stages.

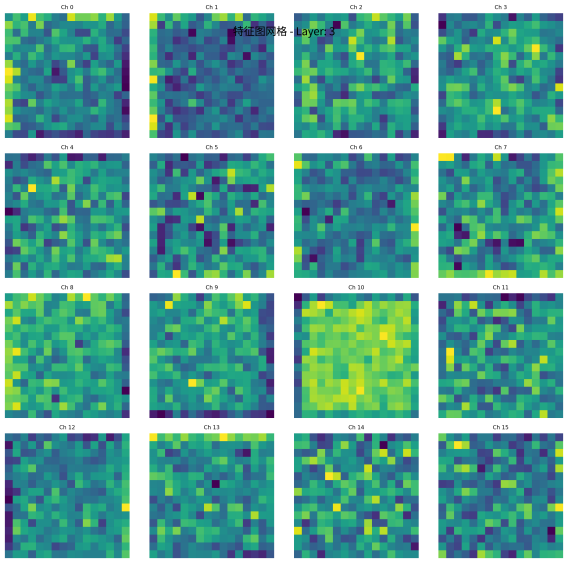
Configuration	**YOLO 10c**	**YOLO 1c**	**EffNet-B0**
Epochs	60	30	10
Image Size ($imgsz$)	512	512	224
Batch Size	64	64	64
Optimizer	AdamW	AdamW	Adam
Initial LR ($lr0$)	0.001	0.001	0.001
Weight Decay	0.0001	0.0001	-
LR Scheduler	Cosine	Cosine	-
Others			
Number parameters	∼3.2M	∼3.2M	∼4M

4.5. Further Analysis of Model Internals and Interpretability

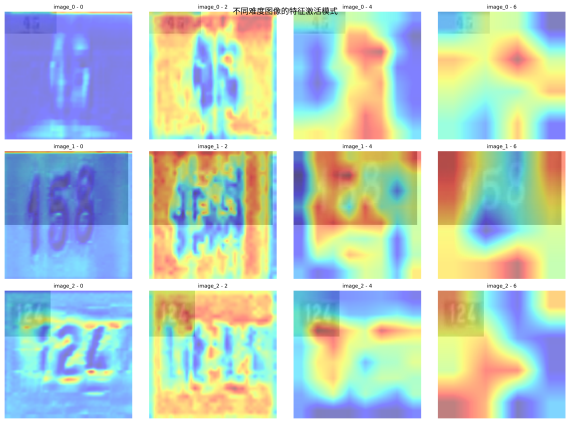
Now we present a qualitative analysis by visualizing the internal representations of our models. Figure 5a displays

some feature maps of our classifier. Each map highlights different spatial regions of an input image, proving that the network has actually learnt a variety of distinct features.

Additionally, Figure 5b shows how activation patterns evolve across layers. Initially, in shallower layers, activations are scattered and digit-specific, corresponding to localized features such as edges or textures. As the network deepens, these patterns converge toward more holistic representations. This shift illustrates the network's increasing capacity to discard irrelevant details and extract higher-level, discriminative traits that can generalize across instances.



(a) EfficientNet-B0 Layer3 feuture maps.



(b) Progression of feature activation patterns.

Figure 5. Internal representations

5. Conclusion and future work

In this work, we have presented a robust two-stage pipeline for Street-View Character Recognition, successfully transitioning from a monolithic multi-head architecture to a decoupled "divide and conquer" strategy. Our core contribution lies in the implementation of a **Semantic-to-Spatial Transfer Learning** approach, where a 10-class YOLOv8 baseline was refined into a high-recall, class-agnostic digit locator.

This modularity proved essential for addressing the inherent challenges of the SVHN dataset. By isolating localization from recognition, we achieved a significant **Recall of 91.99%** in detection. Furthermore, the subsequent specialized classification via EfficientNet-B0 resulted in an overall **Test Accuracy of 91.49%**.

From a computational standpoint, our hybrid framework offers a system that is both lightweight and fast, making it viable for scenarios with limited computational resources. Moreover, the decoupled training significantly reduces convergence time, as each stage optimizes for a specific objective without gradient interference.

Finally, we reached a test accuracy of 91.49% test accuracy, being the 70th position on the ranking of *tian chi competition* [1]. As a future work, we could probably improve this result by diving into data augmentation and hyperparameter tuning, or even trying more complex network architectures.

Table 3. Evolution of Test Accuracy across architectural iterations.

Model Architecture	Strategy	Test Accuracy	Outcome
EfficientNet-B0	Multi-head (Direct)	$\leq 50\%$	Suboptimal
YOLOv8n (Baseline)	10-head Detection	89.73%	Improved
YOLO + EffNet	Hybrid Pipeline	90.83%	+1.10%
YOLO + EffNet	**Hybrid + Transfer***	**91.49%**	**Best (+0.66%)**

*Transfer stands for the proposed Semantic-to-Spatial Transfer Learning.

References

[1] Alibaba Cloud Tianchi. Street view digit recognition challenge. https://tianchi.aliyun.com/competition/entrance/531795/introduction, 2025. 1, 4